

HOW TO SET UP A SOLAR FARM?

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With abundant sunlight 300 days in a year, cost advantage in the production of solar cells and modules, and the impetus provided by the rolling out of Jawaharlal Nehru National Solar Mission (JNNSM) by the central government and followed upon by many state governments, India today has a huge opportunity to become a solar super power. Attractive tariffs being offered by the Central Electricity Regulatory Commission (CERC) for grid-connected solar projects are the prime attraction for setting up solar

photovoltaic (PV) farms. Well-defined government policies are in place. Also, one has access to technically proven and commercially viable technologies, well versed suppliers and engineering procurement and construction contractors, and expert consultants.

Many states have identified certain areas for setting up solar power projects. Hot destinations are Rajasthan, Gujarat, Maharashtra, Madhya Pradesh, Andhra Pradesh, Karnataka and Tamil Nadu as far as solar radiation is concerned. Presently, only Rajasthan and Gujarat have a clear policy and guidelines on the procedures for setting up solar power projects. Other states

may also initiate the policy and procedures soon now that JNNSM has been launched by the government of India.

Payback period for a solar farm

The payback is dependent upon many financial parameters like the cost of debt, depreciation, the capacity utilisation factor, which, in turn, is dependent on the solar resource at the site and the technology adopted (c-Si or thin-film solar), tariff, etc. Using standard numbers and assuming the technology used will be crystalline silicon, under the National Solar Mission incentive structure, the payback period for a solar PV power plant is approximately six years. That is, the cash flows will recover the equity capital in six years though the loans could take longer to clear. Return on equity pre-tax will be about 20 per cent. Project's internal rate of return is about 15.5 per cent.

Eligibility

"Bidding for power projects so far has been technology-neutral. There is no filter put in terms of the experience of the bidder. One, though, needs to have a strong balance sheet, given the banks are reluctant to fund projects. Putting a technical filter will help getting only the experienced and competent companies to get into the business. The government has increased the project size to 5-50 MW for corporates willing to set up the plant. It's a welcome step as it will evoke the interest amongst big, independent power companies like Jindal, Adani and Tata Power,"